
Cloudcasting experiments

Abstract

This document describes the research background, references for this tutorial and summarises the results.

1 Introduction

Cloud forecasting from satellite imagery is an important problem in weather nowcasting, with applications in aviation, renewable energy management, severe weather monitoring, and numerical weather prediction. Accurate short-term forecasts of cloud evolution can provide valuable information about future atmospheric conditions while complementing traditional forecasting systems. Historically, cloud motion has been estimated using optical flow techniques and physics-based numerical weather prediction (NWP) models. More recently, advances in machine learning have enabled data-driven approaches that learn spatiotemporal patterns directly from large archives of satellite observations.

In this tutorial, we explore a sequence of deep learning architectures for satellite image prediction, beginning with recurrent neural networks and progressing toward more advanced sequence modelling approaches. In particular, we investigate Long Short-Term Memory (LSTM) networks, Bidirectional LSTMs, attention mechanisms, and ConvLSTM architectures for forecasting future satellite images from historical observations. Throughout the tutorial, these models are compared against strong baseline approaches including persistence forecasting and optical flow.

The tutorial is based on the Cloudcasting project developed by the Alan Turing Institute.

The Cloudcasting project provides a large-scale benchmark dataset and evaluation framework for cloud forecasting from geostationary satellite observations. The repository contains data processing pipelines, baseline forecasting methods, and benchmark datasets designed to support the development and evaluation of machine learning models for cloud nowcasting.

Unlike many computer vision datasets, satellite imagery represents a spatiotemporal physical process governed by atmospheric dynamics. Consequently, forecasting models must learn both spatial structure and temporal evolution. Clouds undergo complex processes including advection, deformation, growth, and dissipation across multiple spatial and temporal scales. Recurrent neural networks are therefore a natural starting point because they maintain memory of previous observations and can learn temporal dependencies within image sequences.

In this tutorial, students will learn how to process satellite imagery stored in Zarr format, construct forecasting datasets from sequences of satellite observations, implement LSTM-based forecasting models in PyTorch, and evaluate their performance on future-image prediction tasks. The tutorial also introduces modern concepts such as residual forecasting, attention mechanisms, and spatiotemporal sequence modelling that underpin many contemporary weather forecasting systems.

1.1 Data Overview

The Cloudcasting dataset consists of a large archive of geostationary satellite imagery collected from the EUMETSAT Spinning Enhanced Visible and Infrared Imager (SEVIRI) Rapid Scanning Service. The service acquires observations every five minutes over Great Britain and a much larger surrounding region.

The dataset contains observations from eleven SEVIRI spectral channels spanning visible, infrared, and water-vapour wavelengths. These channels provide complementary information about cloud structure, cloud-top temperature, atmospheric moisture, and surface conditions. Over Great Britain, the native spatial resolution corresponds to approximately 4.6×7.1 km per pixel.

For model development, the original five-minute observations are resampled to a temporal resolution of fifteen minutes. The forecasting task is formulated as a three-hour nowcasting problem, where a model receives a sequence of historical satellite images and predicts future cloud conditions. All Cloudcasting benchmark models are trained using approximately three years of historical observations.

1.2 Data Cleaning and Preprocessing

A dedicated preprocessing pipeline was developed to prepare the satellite observations for machine learning applications.

Although SEVIRI provides twelve spectral channels, only eleven channels are retained in the Cloudcasting dataset. The High Resolution Visible (HRV) channel is excluded because its spatial resolution differs from the remaining channels. While the standard SEVIRI channels have a spatial resolution of approximately 2 km per pixel, the HRV channel has a spatial resolution of approximately 1 km per pixel. Excluding the HRV channel preserves a consistent spatial grid across all input variables.

The geographic domain is cropped before download to focus on the United Kingdom and surrounding regions. The resulting spatial extent spans approximately:

- 45°N to 70°N latitude
- 16°W to 10°E longitude

Machine learning models require continuous temporal sequences to learn meaningful spatiotemporal relationships. Therefore, any sequences containing missing observations within either the historical input period or forecasting horizon are removed from the dataset. This ensures that every training example contains a complete and uninterrupted sequence of satellite images.

1.3 Dataset Split

To enable fair comparison between forecasting models, the Cloudcasting dataset is divided into separate training, testing, and verification periods.

The training dataset consists of:

- All observations from 2019–2021
- Alternate fortnights throughout 2022

All model parameters are learned exclusively from this training data.

Two independent evaluation datasets are then used. The remaining alternate fortnights in 2022 form the test dataset. This dataset is used during model development to evaluate forecasting performance and compare different modelling approaches. All observations from 2023 are reserved as a final hold-out verification dataset. This dataset remains unseen during model development and provides an unbiased estimate of how well forecasting models generalise to future weather conditions. This evaluation protocol ensures that models are tested on entirely unseen time periods, providing a realistic assessment of operational forecasting performance.

2 Related Work

Short-term weather forecasting and cloud nowcasting can be formulated as a spatiotemporal sequence prediction problem, where the goal is to predict future atmospheric states from a sequence of historical observations. Traditional approaches rely on numerical weather prediction (NWP) models that explicitly solve the governing equations of atmospheric dynamics. While physically grounded, these methods are computationally expensive and often struggle at short forecasting horizons where

rapidly evolving cloud structures increase forecast uncertainty. Consequently, a growing body of work has explored data-driven approaches based on computer vision and deep learning.

2.1 Recurrent Neural Networks for Spatiotemporal Forecasting

A foundational contribution in this area is the Convolutional LSTM (ConvLSTM) architecture of Shi et al. Shi et al. (2015), which extended traditional LSTM networks by replacing fully connected state transitions with convolutional operations. This modification preserves spatial structure while modelling temporal dependencies, making ConvLSTM particularly suitable for precipitation nowcasting and satellite image forecasting. ConvLSTM demonstrated substantial improvements over classical optical flow methods and established a widely adopted baseline for spatiotemporal forecasting.

Subsequent work improved the ability of recurrent architectures to model complex motion patterns. TrajGRU Shi et al. (2017) introduced dynamically learned recurrent connections that adapt to local motion fields rather than relying on fixed convolutional neighbourhoods. This flexibility enables more accurate modelling of non-uniform motion and deformation, which are common in atmospheric systems. In parallel, PredRNN Wang et al. (2017) proposed a spatiotemporal memory mechanism that explicitly propagates memory states across both temporal and network-depth dimensions. By improving information flow through long sequences, PredRNN demonstrated superior performance on video prediction benchmarks and inspired numerous variants for geospatial forecasting tasks.

Despite their success, recurrent architectures exhibit several limitations. Forecast quality often deteriorates for long prediction horizons due to accumulated error propagation, and recurrent computation limits parallelization during training. These challenges motivated a shift toward attention-based architectures.

2.2 Convolutional Encoder–Decoder Approaches

Alongside recurrent models, encoder–decoder architectures based on convolutional neural networks have emerged as strong baselines for weather forecasting. U-Net Ronneberger et al. (2015) and related fully convolutional networks exploit multi-scale feature extraction and skip connections to preserve spatial detail during image reconstruction. Models such as RainNet Ayzel (2020) demonstrated that carefully designed convolutional architectures can achieve competitive precipitation nowcasting performance without explicit recurrent components.

These approaches efficiently capture hierarchical spatial structure. However, purely convolutional models typically require temporal information to be encoded either through stacked input frames or auxiliary temporal modules, motivating hybrid architectures that combine convolutional feature extraction with recurrent or attention-based sequence modelling.

Large-Scale Attention-Based Forecasting One of the earliest large-scale neural weather forecasting systems was MetNet Sønderby et al. (2020). Unlike earlier ConvLSTM-based approaches that focused primarily on local precipitation nowcasting, MetNet was designed to produce probabilistic precipitation forecasts over continental-scale regions. The model combines radar observations, satellite imagery, topographic information, and forecast lead time to generate probabilistic precipitation maps.

Architecturally, MetNet employs a hybrid design consisting of convolutional feature extraction, ConvLSTM-based temporal encoding, and axial self-attention layers. The ConvLSTM module captures local temporal dynamics from historical observations, while the axial attention mechanism aggregates long-range spatial dependencies across input regions spanning more than one million square kilometres. By decomposing two-dimensional attention into separate horizontal and vertical attention operations, MetNet achieves substantially lower computational complexity than standard self-attention while maintaining a large receptive field.

Experimental results demonstrated that MetNet outperformed operational numerical weather prediction systems for short-range precipitation forecasting horizons of up to 7–8 hours over the continental United States. These results provided early evidence that deep neural architectures could effectively complement or surpass traditional forecasting systems in the nowcasting regime.

MetNet-2 Espeholt et al. (2021) extended this paradigm by incorporating substantially larger spatial contexts and probabilistic forecasting objectives. The model demonstrated further improvements in

precipitation forecasting skill and highlighted the importance of large-context neural architectures for atmospheric prediction.

2.3 Generative Approaches

A persistent challenge in weather forecasting is the inherent uncertainty of atmospheric evolution. Deterministic models often produce overly smooth predictions that fail to capture realistic variability, especially at longer lead times. To address this issue, recent work has explored probabilistic generative approaches.

DeepMind’s DGMR Ravuri et al. (2021) introduced a deep generative model for radar nowcasting capable of producing multiple plausible future precipitation scenarios. Rather than predicting a single deterministic forecast, DGMR learns a distribution over future atmospheric states, significantly improving realism and uncertainty quantification.

More recently, diffusion-based approaches have emerged as a powerful paradigm for geophysical forecasting. Models such as PreDiff Gao et al. (2023) apply latent diffusion processes to precipitation nowcasting, generating sharper and more physically plausible forecasts than traditional deterministic models. Similar diffusion-based architectures have subsequently been applied to satellite cloud forecasting and geostationary infrared imagery, demonstrating promising results for probabilistic cloud evolution modelling.

2.4 Graph Neural Networks for Weather Forecasting

Recent advances in machine learning for weather prediction have moved beyond convolutional and recurrent architectures toward graph-based representations of atmospheric dynamics. A notable example is GraphCast Lam et al. (2023), which demonstrated that graph neural networks (GNNs) can achieve state-of-the-art medium-range weather forecasting performance while operating orders of magnitude faster than traditional numerical weather prediction systems.

GraphCast formulates weather forecasting as an autoregressive prediction problem. Given atmospheric states at the current time and six hours earlier, the model predicts the next atmospheric state six hours into the future. Forecasts at longer lead times are obtained by recursively feeding model predictions back into the network. The atmospheric state is represented on a global latitude–longitude grid at 0.25° resolution, corresponding to approximately one million spatial locations and over two hundred meteorological variables per grid point.

Unlike convolutional architectures that operate directly on regular grids, GraphCast employs an encoder–processor–decoder graph neural network architecture. The encoder maps atmospheric variables from the latitude–longitude grid to an internal multimesh graph representation. This multimesh is derived from a hierarchy of refined icosahedral meshes projected onto the sphere, producing a nearly uniform spatial discretization of the Earth’s surface. The resulting graph contains approximately 40,000 nodes and supports efficient information propagation across local and global spatial scales.

The processor consists of multiple graph neural network layers that perform learned message passing across the multimesh. Through successive message-passing operations, information can propagate over large geographic distances, enabling the model to capture both local atmospheric interactions and planetary-scale circulation patterns. Finally, the decoder projects the learned graph representation back onto the latitude–longitude grid and predicts the future atmospheric state.

A particularly important design choice is the use of residual prediction. Rather than forecasting the full atmospheric state directly, GraphCast predicts an update relative to the most recent input state. This residual formulation improves optimization and reflects the fact that atmospheric variables often evolve smoothly between consecutive forecasting intervals.

GraphCast represents a significant conceptual shift in machine learning weather forecasting. Whereas ConvLSTM and Transformer-based nowcasting models primarily focus on local spatiotemporal evolution within image-like representations, GraphCast explicitly exploits the spherical geometry of the Earth and learns atmospheric dynamics through graph-based message passing. This allows efficient modelling of long-range interactions that are difficult to capture using conventional convolutional

architectures. The success of GraphCast demonstrates the potential of graph neural networks as a foundation for future Earth-system forecasting models.

2.5 Satellite Cloud Forecasting

While many forecasting studies focus on radar-based precipitation nowcasting, a growing literature specifically addresses cloud forecasting from satellite observations. Satellite imagery presents distinct challenges due to the large spatial extent of observations, multispectral sensing modalities, and the complex dynamics of cloud formation and dissipation.

Recent systems such as CloudCast Partio et al. (2025) and SATcast Chen et al. (2025) leverage modern deep learning architectures to forecast future cloud fields directly from geostationary satellite observations. These approaches combine advances in convolutional networks, attention mechanisms, and generative modelling to capture both local cloud dynamics and large-scale atmospheric structure. The increasing availability of high-resolution satellite archives and cloud-native data formats such as Zarr has further accelerated research in this area.

2.6 Positioning of the Present Work

The progression from ConvLSTM to attention-based and generative architectures reflects a broader trend in spatiotemporal forecasting toward models capable of capturing increasingly complex atmospheric dynamics. Nevertheless, ConvLSTM remains an important baseline due to its computational efficiency, interpretability, and strong performance on short-term forecasting tasks.

In this work, we investigate a hierarchy of forecasting models beginning with persistence and optical-flow baselines, followed by recurrent architectures including ConvLSTM and Bidirectional LSTM networks, and extending to attention-enhanced sequence models. This progression enables a systematic evaluation of how increasingly expressive temporal representations influence cloud forecasting performance on satellite imagery.

3 Methods

3.1 Optical Flow

Optical flow provides a physics-inspired baseline for satellite image forecasting by estimating the apparent motion of cloud structures between consecutive images. Under the assumptions of brightness constancy and spatially coherent motion, optical flow computes a dense displacement field that describes how pixels move from one frame to the next. In this work, two widely used optical flow methods were evaluated: Farnebäck Dense Optical Flow (Farnebäck, 2003) and TVL1 Optical Flow (Sánchez Pérez et al., 2013; Zach et al., 2007). Farnebäck’s method estimates motion by modelling local image neighbourhoods with polynomial expansions, whereas TVL1 combines an L^1 data fidelity term with total variation regularisation, providing greater robustness to noise and motion discontinuities. Both methods were implemented using OpenCV (Bradski, 2000) and applied independently to each spectral channel. Given satellite images at times $t - 1$ and t , optical flow vectors were computed and used to generate forecasts at $t + 1$. Multi-step forecasts up to a 180-minute horizon were obtained autoregressively by repeatedly feeding each newly predicted image back into the optical flow model, thereby providing strong non-learning baselines for comparison with deep learning approaches.

3.2 RAFT Optical Flow

As a deep learning alternative to classical optical flow methods, we evaluate the Recurrent All-Pairs Field Transforms (RAFT) model (Teed and Deng, 2020). RAFT is a state-of-the-art optical flow architecture that estimates dense pixel-wise motion between two images using a recurrent refinement process. Unlike traditional optical flow algorithms, RAFT constructs a high-resolution all-pairs correlation volume that captures similarities between every pixel in the first image and every pixel in the second image. A recurrent update operator then iteratively refines the flow field over multiple steps, enabling highly accurate motion estimation. In this work, we use the pretrained RAFT implementation available in TorchVision. Given two consecutive satellite images at times $t - 1$ and t ,

RAFT predicts a dense motion field describing cloud displacement between the frames. This flow field is subsequently used to advect the most recent satellite image and generate a forecast for the next timestep. Although RAFT was originally trained on natural-image datasets rather than satellite observations, it provides a strong deep-learning baseline for comparison with classical optical flow methods and recurrent forecasting models.

3.3 LSTM-Based Forecasting Experiments

To evaluate recurrent neural network architectures for short-term cloud forecasting, a series of experiments were conducted using the VIS008 visible satellite channel. The forecasting task consisted of predicting the next satellite image at a 15-minute lead time using the previous five observations as input. All models were trained on daytime imagery using downsampled satellite images within a CPU-based notebook environment.

Several recurrent architectures were investigated, including a standard ConvLSTM, residual ConvLSTM variants, stacked ConvLSTM networks, and a CNN-BiLSTM encoder-decoder model. The plain ConvLSTM successfully learned temporal patterns in the training data but failed to outperform the persistence baseline. This result is consistent with the nature of short-horizon satellite forecasting, where the most recent observation often provides a strong approximation of the next frame.

To address this limitation, a residual forecasting strategy was adopted. Instead of directly predicting the future image, the model was trained to predict the residual change relative to the most recent observation. This formulation significantly improved forecasting performance by allowing the network to focus on cloud motion and evolution rather than reconstructing the entire image. The best-performing model consisted of a single-layer residual ConvLSTM with 32 hidden channels operating on images of size 128×208 . The model was trained using a masked loss function and a combined mean squared error (MSE) and mean absolute error (L1) objective.

Additional experiments explored two architectural extensions. First, a sharper residual ConvLSTM was trained using an additional gradient-based loss designed to preserve image sharpness and cloud boundaries. Second, a stacked residual ConvLSTM with multiple recurrent layers was evaluated to increase model capacity. Both approaches improved performance relative to persistence and reduced large forecasting errors, particularly in the upper tail of the error distribution. However, neither architecture exceeded the performance of the simpler single-layer residual ConvLSTM.

A CNN-BiLSTM encoder-decoder architecture was also evaluated. In this approach, convolutional layers compressed each image into a compact feature representation before temporal processing by a bidirectional LSTM. The model performed substantially worse than all ConvLSTM variants, suggesting that compressing entire satellite images into low-dimensional feature vectors results in the loss of important spatial information required for accurate cloud forecasting.

Overall, the experiments demonstrate the importance of preserving spatial structure throughout the forecasting pipeline. Among all architectures considered, the residual ConvLSTM provided the most favourable balance between model complexity, computational efficiency, and forecasting accuracy. The results further highlight the effectiveness of residual forecasting, where the network learns corrections to a strong persistence baseline rather than attempting to reconstruct future satellite imagery from scratch.

4 Results

In the LSTM notebook, we trained several neural-network baselines for next-frame satellite image prediction using the VIS008 visible channel. The task was to use the previous five satellite frames to predict the next frame at a 15-minute lead time. All models were trained on daytime imagery, using downsampled images and a notebook-safe CPU setup.

The plain ConvLSTM successfully learned the training data, but its predictions were still blurry and did not outperform the persistence baseline. This was expected: for 15-minute forecasting, persistence is a very strong baseline because the next satellite image is often very similar to the most recent image.

The best improvement came from using a residual ConvLSTM. Instead of predicting the next image from scratch, the model predicted a correction to the last observed frame. This made the learning

problem easier and allowed the model to beat persistence. Increasing the hidden size to 32, using a larger resolution of 128×208 , applying a masked loss, and training with a combined MSE + L1 objective produced the best result of the day.

We also tested a sharper residual ConvLSTM with an additional gradient loss, and a stacked residual ConvLSTM with two recurrent layers. Both models beat persistence in masked MSE and reduced large errors, especially in the 95th and 99th percentile error ranges. However, neither improved over the best single-layer residual ConvLSTM. The CNN + BiLSTM model performed poorly, confirming that compressing each satellite image into a single vector loses too much spatial information for this task.

5 Performance Summary

Model	Masked MSE	MAE	95th Perc.	99th Perc.
Persistence baseline	0.001039	0.01647	0.06360	0.11591
Plain ConvLSTM	0.001637	—	—	—
Improved Residual ConvLSTM	0.000838	0.01579	0.05522	0.10003
Sharp residual ConvLSTM	0.000900	0.01707	0.05728	0.10230
Stacked residual ConvLSTM	0.000891	0.01703	0.05717	0.10290
CNN + BiLSTM	0.01870	—	—	—

Table 1: Performance comparison of next-frame satellite image prediction models.

6 Main Conclusion

The best model so far is the improved residual ConvLSTM. It is the only model that improved over persistence across all recorded metrics. The residual design is important because it starts from the strong persistence baseline and learns only the change needed to predict the next frame. For satellite nowcasting, this is a much better approach than asking the model to reconstruct the full image from scratch.

The experiments also showed that more complexity does not automatically improve performance. The stacked ConvLSTM and gradient-loss model helped reduce large errors, but they introduced more small errors across the image. For the current dataset size and CPU-based notebook workflow, the simpler single-layer residual ConvLSTM is the strongest and most reliable model.

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